Programming Fundamentals

Assignment 3 – Part 1 | Solutions

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Problem 1: Average in an Array

Approach: Store N scores in an array, pass to calculateAverage() which sums all elements and divides by size. Display result with 2 decimal places.

C++ Code

```
#include <iostream>
using namespace std;

double calculateAverage(int arr[], int size) {
    double sum = 0;
    for (int i = 0; i < size; i++) {
        sum += arr[i];
    }
    return sum / size;
}

int main() {
    int n;
    cout << "Enter number of students: ";
    cin >> n;
    int scores[n];
    cout << "Enter scores: ";
    for (int i = 0; i < n; i++) cin >> scores[i];
    cout << fixed; cout.precision(2);
    cout << "Average Score: " << calculateAverage(scores, n) << endl;
    return 0;
}
```

Problem 2: Largest and Smallest Values in an Array

Approach: Two separate functions — calculateLowest() and calculateHighest() — each traverse the array tracking a running min/max value initialized to arr[0].

C++ Code

```
#include <iostream>
using namespace std;

int calculateLowest(int arr[], int size) {
    int lowest = arr[0];
    for (int i = 1; i < size; i++) {
        if (arr[i] < lowest) lowest = arr[i];
    }
    return lowest;
}
```

```
int calculateHighest(int arr[], int size) {
    int highest = arr[0];
    for (int i = 1; i < size; i++) {
        if (arr[i] > highest) highest = arr[i];
    }
    return highest;
}

int main() {
    int n;
    cout << "Enter number of students: ";
    cin >> n;
    int scores[n];
    cout << "Enter scores: ";
    for (int i = 0; i < n; i++) cin >> scores[i];
    cout << "Lowest Score: " << calculateLowest(scores, n) << endl;
    cout << "Highest Score: " << calculateHighest(scores, n) << endl;
    return 0;
}
```

Problem 3: Custom Power Function

Approach: calculatePower() uses a loop to multiply base by itself exponent times (starting with result = 1), avoiding the built-in pow() function.

C++ Code

```
#include <iostream>
using namespace std;

int calculatePower(int base, int exponent) {
    int result = 1;
    for (int i = 0; i < exponent; i++) {
        result *= base;
    }
    return result;
}

int main() {
    int base, exponent;
    cout << "Enter base and exponent: ";
    cin >> base >> exponent;
    int result = calculatePower(base, exponent);
    cout << base << " raised to the power " << exponent
         << " is: " << result << endl;
    return 0;
}
```

Problem 4: Voltage Calculator

Approach: calculateVoltage() takes current and resistance as doubles and returns their product (V = I × R) using Ohm's Law.

C++ Code

```
#include <iostream>
using namespace std;

double calculateVoltage(double current, double resistance) {
    return current * resistance;
}

int main() {
    double current, resistance;
    cout << "Enter current and resistance: ";
    cin >> current >> resistance;
    double voltage = calculateVoltage(current, resistance);
    cout << "Total Voltage: " << voltage << " V" << endl;
    return 0;
}
```

Problem 5: Cricket Team Initial Finder

Approach: linearSearch() iterates over every index of the char array. If arr[i] == target, returns true immediately. If the loop ends without a match, returns false.

C++ Code

```
#include <iostream>
using namespace std;

bool linearSearch(char arr[], int size, char target) {
    for (int i = 0; i < size; i++) {
        if (arr[i] == target) return true;
    }
    return false;
}

int main() {
    char team[] = {'b', 'r', 'f', 's', 'n', 'a', 'h', 'i', 'u', 'z', 'm'};
    int size = 11; char target;
    cout << "Enter player initial to search: ";
    cin >> target;
    if (linearSearch(team, size, target))
        cout << "Player is in the team" << endl;
    else
        cout << "Player not found" << endl;
    return 0;
}
```

Problem 6: Word Length Calculator

Approach: wordLength() uses a while loop that increments a counter until the null character '\0' is encountered — replicating strlen() behaviour manually.

C++ Code

```
#include <iostream>
using namespace std;
```

```
int wordLength(char arr[]) {
    int count = 0;
    while (arr[count] != '\0') {
        count++;
    }
    return count;
}

int main() {
    char word[100];
    cout << "Enter a word: ";
    cin >> word;
    cout << "Length of the word is: " << wordLength(word) << endl;
    return 0;
}
```